

APPENDIX

A. Summary of Key Notations

To facilitate understanding, key notations used in this paper are listed in Table VI.

Table VI List of some key notations.

Notation	Description
<i>VirEqPhy</i>	The proposed method for geometric dynamic equivalence between virtual and physical spaces which mainly includes multidimensional feature extraction module MI , voxel representation learning module MV , model embedding retrieval module MR and model cross-domain alignment module MA .
<i>I_m</i>, <i>DS</i>	RGB image that describes the real scene observed by the agent and digital space corresponding to the real scene.
<i>E</i>, <i>DB</i>	Set of objects $\{obj_1, \dots, obj_{ E }\}$ in the real scene and CAD model library $\{cad_1, \dots, cad_{ DB }\}$ of objects.
<i>F_I</i>, <i>F_V</i>	Image representation of <i>I_m</i> and voxel representation of objects. The former includes image depth <i>D</i> , object class <i>c</i> , bounding box <i>b</i> , and mask <i>m</i> . The latter contains 3D voxel grid v_{obj} and voxel coordinates n_{obj} of objects.
<i>s</i>, <i>t</i>, <i>r</i>	Scale, position, and rotation of objects.
<i>SM</i>^{sg}_{gt}	Normalized similarity matrix of all objects and CAD models in weakly supervised datasets based on appearance similarity S_{gt}^{app} , semantic similarity S_{gt}^{sem} , and geometric similarity S_{gt}^{geo} . Each similarity is calculated separately through appearance feature f^{app} , semantic feature f^{sem} , and voxel feature f^{vox} .
<i>SM</i>, <i>I</i>	Normalized similarity matrix and its indicator matrix between objects and CAD models.

B. Supplementary Qualitative Results for *VirEqPhy*

Fig. 12 further visualizes inference results for occluded objects. Although some detail loss occurs, our method's predicted voxel representations retain richer visual information than 2D features, enabling retrieval of the most similar CAD models.

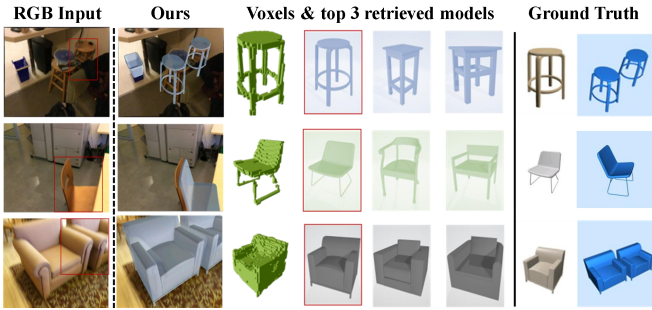


Fig. 12. Qualitative results in scenarios with partially occluded objects.

C. Supplementary Qualitative Results for **W-MR**

Fig. 13 provides a *t*-SNE visualization of the embedding space after training, illustrating how different methods group CAD models. From the figure, it can be seen that **W-MR** forms tighter intra-class clusters and clearer inter-class separations than the baselines, indicating a more discriminative retrieval space. This enables the retrieved models to exhibit more precise separation between intra-class and inter-class instances. This qualitative result aligns with Table V, clearly indicating that the **W-MR** can enhance the model retrieval ability.

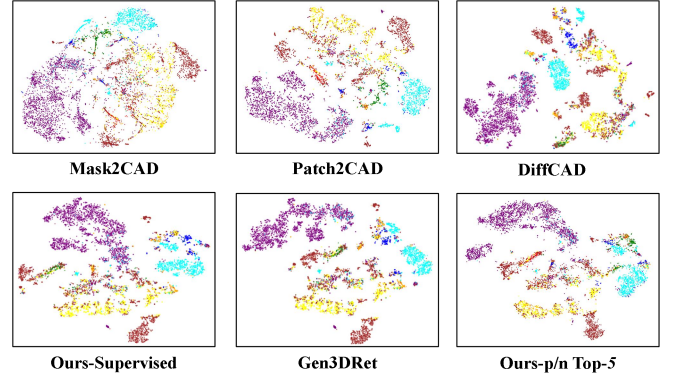
Fig. 13. *t*-SNE visualization of embedding space of CAD models. The same color represents the same class of CAD models.

Fig. 14 visualizes the most similar CAD models (intra-class) and least similar CAD models (inter-class) retrieved by **Our-p/n Top-1** for unseen objects during training. The most similar CAD models exhibit nearly identical geometric and semantic characteristics to target objects, while the least similar models (used as negative examples) belong to distinct categories or have dissimilar shapes. This confirms that our method with positive/negative samples not only identifies optimal matches but also learns dissimilar boundaries.

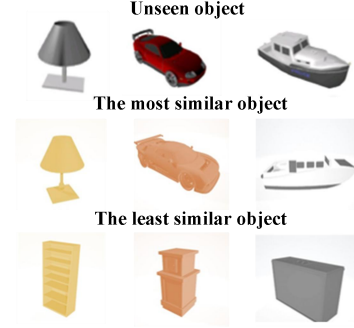


Fig. 14. Most similar and least similar CAD models of some unseen objects.

A dynamic visualization in Fig. 15 shows that, as the camera moves, our method consistently retrieves the correct CAD models and refines 9-DoF poses in real time. In conclusion, this CAD model retrieval enhancement module reduces manual annotation cost while improving *VirEqPhy*'s robustness and accuracy for unseen object. This generalization ability is essential for expanding digital twin systems, enabling adaptation to dynamic environments.

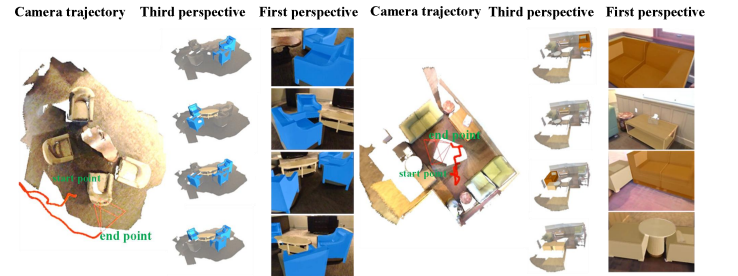


Fig. 15. Dynamic visualization results.

D. Supplementary Qualitative Results for Real-World Agents

Fig. 16 shows virtual scene updates under dynamic objects layouts. Similarly, our method promptly captures object motion in the physical space (e.g., moving chair) and synchronize it in the virtual space without perceptible delay or model drift. This efficiency stems from the lightweight multi-task framework and highlights the system’s real-time updating capability, which is critical for embodied tasks such as navigation, obstacle avoidance, and dynamic interaction with moving objects.



Fig. 16. Virtual space updates in dynamic scene.

E. Performance of Cross-Domain Transfer

To further evaluate the cross-domain transferability of the proposed *VirEqPhy*, we conduct an additional experiment on the high-fidelity simulation platform AI Habitat. It is worth noting that cross-domain evaluation in AI Habitat offers several advantages: it enables controllable, reproducible shifts (lighting, layout, object mix), supplies precise annotations (RGB/depth/semantic and 9-DoF), and provides low-cost scalability to add new CADs for efficient multi-scene generalization tests.



Fig. 17. Different types of light sources and examples of the new CAD models.

1) Experimental Settings We create 60 virtual indoor scenes by introducing various light sources, additional CAD models (Fig. 17), and randomly distributing small objects (e.g., bowls, books) to increase scene complexity. Each scene is captured by a pinhole camera mounted on the agent, recording RGB, depth, and semantic images under different lighting conditions, resulting in 2,400 data samples. *VirEqPhy* is trained on the Scan2CAD-based dataset (i.e., ScanNet-ShapeNet alignment) and then evaluated on synthetic scenes created in AI Habitat.

2) Results The results in Table VII show that all methods perform better on the synthetic dataset, primarily due to more balanced object distributions and controlled lighting and layout. Our method achieves state-of-the-art performance on both class-level and instance-level metrics, indicating robust

generalization under shifts in lighting, layout, and object composition. We attribute this improvement to preserving fine-grained geometric cues and tightly coupling retrieval with alignment in a unified framework.

Table VII Retrieval-Aware Alignment Accuracy (%) on different datasets.

Method	Scan2CAD-based dataset		Habitat-based synthetic dataset	
	class	instance	class	instance
Mask2CAD*	9.5	13.8	18.6	20.0
ROCA*	17.6	21.7	23.2	25.1
DiffCAD*	19.8	24.6	24.8	27.0
FastCAD*	18.5	21.4	23.2	25.0
InCAD*	19.4	23.0	24.5	26.4
Ours	20.6	24.5	25.2	27.8

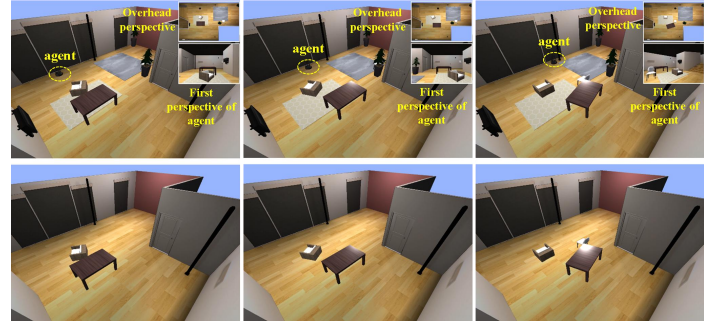


Fig. 18. Top: Scene changes in Habitat, Bottom: Scene updating based on the proposed method.

Fig. 18 illustrates the scene dynamics and the corresponding digital-twin updates during continuous indoor navigation. Despite viewpoint changes, occlusion by tables and sofas, and the introduction of a new object (a chair), the system consistently retrieves the correct CAD models and maintains accurate 9-DoF alignment. These results further demonstrate effective cross-domain transfer beyond the original data domain.

F. Case Study on Multi-Agent Restaurant Service

We conditionally deploy the proposed *VirEqPhy* in a restaurant-service scenario, where multiple heterogeneous agents autonomously navigate and collaboratively deliver, place, and retrieve tableware and food items of varying shapes and sizes to enhance the dining experience. For these non-standard objects, we generate CAD models via 3D reconstruction and augment the retrieval database accordingly. Experimental results indicate that *VirEqPhy* tracks geometric changes in the physical space in real time with satisfactory accuracy in this setting. Due to space limitations, additional details are provided in the supplementary material and at <https://forbiddenname.github.io/VirEqPhyDT>.